

Transmission Network Reduction

We merge buses so that a DC-OPF on the smaller network still gives a dispatch that works on the original one. All three approaches choose the same thing — which lines to collapse — and differ in what they require of the result.

1 Notation

$\mathcal{B}, \mathcal{L}, \mathcal{G}$	buses 1.. N , lines 1.. L , generators 1.. K
$u(l), v(l)$	ends of line l ; $n(k)$ the bus of generator k
E	bus–line incidence: $E_{bl} = +1$ if $u(l) = b$, -1 if $v(l) = b$, else 0
C	bus–generator incidence: $C_{bk} = 1$ if $n(k) = b$, else 0
$\mathbf{B}$	$\text{diag}(B_1, \dots, B_L)$, the line susceptances
F_l	thermal rating of line l
$\underline{T}_l, \overline{T}_l$	bounds on the transfer a collapsed line may carry
c_k	linear cost of generator k ; $\underline{g}_k, \overline{g}_k$ its limits
d	demand at each bus; $p = Cg - d$ the nodal injections
g, θ, f	generation, bus angles, line flows
t	internal transfer, carried only by a collapsed line
z_l	1 if line l is collapsed, 0 if it survives
Z^*	optimal cost of the full network (a constant)

2 What a reduction is

Both MILP models express the reduction on the original graph, using one switch:

$$-F_l(1 - z_l) \leq f_l \leq F_l(1 - z_l) \quad \text{switched limit} \quad (1)$$

At $z_l = 0$ this is the ordinary limit. At $z_l = 1$ it forces $f_l = 0$, hence $\theta_{u(l)} = \theta_{v(l)}$: the buses merge. The power the line carries is then taken up by the internal transfer t_l , which is free only on a collapsed line. So a collapsed line loses its thermal limit, which is the whole difficulty.

With the transfer present the DC power flow reads $E \mathbf{B} E^\top \theta + Et = Cg - d$.

3 Flow matching

The **proxy/** approach. Fix the injections at the values the full OPF produced and ask the reduced network to reproduce the same flows to within a tolerance. No optimisation sits inside the model, so it is one MILP. Here $\hat{f}$ and p are the full network's optimal flows and injections, and ε_l is the window half-width (**eps** times the rating). Lines loaded above **near_limit** are held out of any cluster.

$$\max_{z, \theta, f, t} \sum_{l \in \mathcal{L}} z_l \quad \text{maximise merges} \quad (2)$$

$$f = \mathbf{B} E^\top \theta \quad \text{line flows} \quad (3)$$

$$E(f + t) = p \quad \text{DC power flow, fixed injections} \quad (4)$$

$$(\hat{f}_l - \varepsilon_l)(1 - z_l) \leq f_l \leq (\hat{f}_l + \varepsilon_l)(1 - z_l) \quad \text{flow window, switched} \quad (5)$$

$$\underline{T}_l z_l \leq t_l \leq \overline{T}_l z_l \quad \text{internal transfer} \quad (6)$$

$$z_l = 0, \quad l \text{ congested} \quad \text{line stays unmerged} \quad (7)$$

for all $l \in \mathcal{L}$.

The injections are fixed, so accuracy is asserted only at the dispatch the full network already chose. A DC-OPF on the reduced network may pick a different one, and that dispatch can overload the original network.

4 Bilevel

Ask for what flow matching cannot: the dispatch the *reduced* network returns must be deliverable on the original one. Here $\tilde{\theta}, \tilde{f}$ are the angles and flows the original network carries under that dispatch.

$$\max_{z \in \{0,1\}^L} \sum_{l \in \mathcal{L}} z_l \quad \text{maximise merges} \quad (8)$$

$$\tilde{f} = \mathbf{B} E^\top \tilde{\theta}, \quad |\tilde{f}_l| \leq F_l \quad \text{deliverable on full network} \quad (9)$$

$$E \mathbf{B} E^\top \tilde{\theta} = Cg - d \quad \text{DC power flow, full network} \quad (10)$$

$$c^\top g \leq (1 + \frac{\gamma}{100}) Z^\star \quad \text{cost within tolerance} \quad (11)$$

$$(g, \theta, f, t) \in \arg \min \text{LL}(z) \quad \text{lower level optimal} \quad (12)$$

The lower level $\text{LL}(z)$ is the reduced network's DC-OPF:

$$\min_{g, \theta, f, t} c^\top g \quad \text{minimise cost} \quad (13)$$

$$f = \mathbf{B} E^\top \theta \quad \text{line flows} \quad (14)$$

$$E(f + t) = Cg - d \quad \text{DC power flow} \quad (15)$$

$$-F_l(1 - z_l) \leq f_l \leq F_l(1 - z_l) \quad \text{switched limit} \quad (16)$$

$$\underline{T}_l z_l \leq t_l \leq \overline{T}_l z_l \quad \text{internal transfer} \quad (17)$$

$$\underline{g}_k \leq g_k \leq \overline{g}_k \quad \text{generator limits} \quad (18)$$

4.1 Single-level KKT form

$\text{LL}(z)$ is an LP for fixed z , so (12) can be replaced by its KKT conditions. The single-level model is

(8)–(11) (upper level), (13)–(18) (lower-level feasibility), and the rows below.

Duals are λ on (15), σ on (14), $\mu^\pm \geq 0$ on (16), $\tau^\pm \geq 0$ on (17), and $\rho^\pm \geq 0$ on (18).

$$c - C^\top \lambda + \rho^+ - \rho^- = 0 \quad (g) \quad (19)$$

$$E^\top \lambda + \sigma + \mu^+ - \mu^- = 0 \quad (f) \quad (20)$$

$$E^\top \lambda + \tau^+ - \tau^- = 0 \quad (t) \quad (21)$$

$$E \mathbf{B} \sigma = 0 \quad (\theta) \quad (22)$$

$$\mu_l^+ (F_l(1 - z_l) - f_l) = 0 \quad \text{complementarity, flow upper} \quad (23)$$

$$\mu_l^- (F_l(1 - z_l) + f_l) = 0 \quad \text{complementarity, flow lower} \quad (24)$$

$$\tau_l^+ (\overline{T}_l z_l - t_l) = 0 \quad \text{complementarity, transfer upper} \quad (25)$$

$$\tau_l^- (t_l - \underline{T}_l z_l) = 0 \quad \text{complementarity, transfer lower} \quad (26)$$

$$\rho_k^+ (\overline{g}_k - g_k) = 0, \quad \rho_k^- (g_k - \underline{g}_k) = 0 \quad \text{complementarity, generators} \quad (27)$$

5 Greedy

Merge one line at a time, accepting a merge only if two LPs confirm the same condition the bilevel model asks for. Both LPs carry the reduced and the original network at once, sharing the dispatch g .

$$\phi(z) = \min c^\top g \text{ s.t. reduced network} \quad \text{reduced optimum} \quad (28)$$

$$\eta(z) = \min \eta \text{ s.t. } |\tilde{f}_l| \leq F_l(1 + \eta), \quad c^\top g \leq \phi(z) + \epsilon_c \quad \text{cheapest violation} \quad (29)$$

$$\phi(z) \leq \left(1 + \frac{\gamma}{100}\right) Z^* \quad \text{accept if: cost within tolerance} \quad (30)$$

$$\eta(z) \leq \epsilon_f \quad \text{accept if: no overload} \quad (31)$$

and the same must hold at every design demand. Equation (29) searches the reduced network's optimal cost band for a dispatch safe on the original network, which is the “some optimum” of (12).

Candidates are ordered by how much a merge moves the flows. Collapsing line (i, j) at fixed injections shifts every flow by

$$\Delta f = \kappa (H_{:,i} - H_{:,j}), \quad \kappa = -\frac{\theta_i - \theta_j}{X_{ii} + X_{jj} - 2X_{ij}} \quad \text{predicted flow change} \quad (32)$$

$$\text{score} = \alpha \|\Delta f/w\|_2 + (1 - \alpha) \|\Delta f/w\|_\infty \quad \text{ranking score} \quad (33)$$

with H the PTDF, X the reference-removed inverse of $E\mathbf{B}E^\top$, and w the normaliser: either the ratings F or the remaining headroom $F - |f|$, set by `norm`. This is exact; after each accepted merge X and H are updated by rank-one formulas.

Algorithm 1 Greedy reduction

- 1: $z \leftarrow 0$; check the full network passes the acceptance test
 - 2: merge every radial, generator-free line without an LP
 - 3: build X , H and score all candidates
 - 4: **while** time remains and candidates are left **do**
 - 5: $l \leftarrow$ lowest-scoring candidate
 - 6: $z' \leftarrow z$ with l , and every line its clustering makes internal, set to 1
 - 7: **if** z' passes the acceptance test at every design demand **then**
 - 8: $z \leftarrow z'$; update X , H ; re-score
 - 9: **else**
 - 10: drop l
 - 11: **end if**
 - 12: **end while**
 - 13: **optional:** solve the joint KKT model at fixed z ; on failure roll back and repeat
 - 14: **return** z
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Every accepted merge satisfies the bilevel condition. Nothing says the reduction is the largest possible: there is no dual bound, and merges are irreversible, so it can stall while a better reduction exists that needs two changes at once.

6 Hop limit

A cluster is a group of buses tied together by collapsed lines. Nothing so far stops the optimiser from tying together buses that sit far apart, which is how a reduction ends up merging distant parts of the network into one node. The hop limit bounds that reach: inside any cluster, no chain of collapsed lines is longer than k lines.

Let $\mathcal{P}_k$ be the simple paths of the original graph that use exactly $k+1$ lines, and $|P|$ the number of lines in P . Collapsing every line of such a path puts its two ends in one cluster, $k+1$ hops apart. Forbidding that is the whole constraint:

$$\sum_{l \in P} z_l \leq |P| - 1, \quad P \in \mathcal{P}_k \quad \text{path not fully collapsed} \quad (34)$$

Longer chains need no row of their own: a collapsed chain of more than k lines contains $k+1$ consecutive collapsed lines, and that sub-chain is a path in $\mathcal{P}_k$ whose row is already violated. Paths running through a line held out of every cluster are skipped, since such a line has $z_l = 0$ and its row can never be tight.

The limit removes partitions the unrestricted problem allows, so the reduction it reports is not globally optimal.

7 Summary

	injections	guarantee
Flow matching	fixed at the full optimum	none on the re-optimised dispatch
Bilevel	chosen by the reduced OPF	some reduced optimum is safe and cheap
Greedy	chosen by the reduced OPF	same, per accepted merge

All three certify only the demands in the design set, and only for the lossless DC model.